The tail after a critical head is intransitive

Human Compression project: L044 continuation

Retain all acquired critical-head hypotheses and proofs, including the smaller-degree alternating induction used to reach that configuration. Thus $G \leq A_n$ has a transitive connected string C-marking, all full maximal omissions are intransitive, and rank failure has already given $n = 2r \geq 18$. The first three marks form the actual six-point head on D, and with $E = \Omega \setminus D$ one has

$$\rho_3 = (6\ b)\delta, \quad \mathrm{supp}(\delta) \subseteq E \setminus \{b\}, \qquad B = G_{\geq 4}, \quad B|_D = 1.$$

All generators of B except its first fix b.

Theorem 1. *The group B is intransitive on E. Consequently colour 3 has at least two full ambient-eligible edges.*

Proof. Suppose B is transitive. Restriction to E is faithful, since every element fixes D. It therefore transfers the full intersection property, independence and the unchanged nonidentity marks. Its degree and rank are

$$f = |E| = n - 6 \geq 12, \qquad d = r - 4 = f/2 - 1.$$

Every mark is even on E. Its noncommutation diagram is connected, as a terminal interval of the connected full string. Every distinguished omission in B is intransitive on E: otherwise the head and the edge $(6\ b)$ of ρ_3 would make the corresponding full ambient omission transitive.

If B is proper primitive in A_E, the complete accepted primitive bound gives

$$d \leq \left\lfloor \frac{f-3}{2} \right\rfloor = f/2 - 2,$$

contrary to its displayed rank. This uses the full primitive proof and its required finite boundary; it is not an invocation of the alternating rank theorem at degree f. If $B = A_E$ naturally, the accepted pure-head obstruction applies directly, with $i = 3$, the actual head transitive on D, and the literal crossing $(6\ b)\delta$ fixing $D \setminus \{6\}$. It excludes this case without any rank estimate.

It remains that B is imprimitive. Choose any nontrivial block system, of block size $a \geq 2$ and count $z \geq 2$. The accepted two-block theorem excludes $z = 2$ in every critical degree $n \geq 18$. The accepted uniform pair-block theorem excludes $a = 2$ in that same range. If $a, z \geq 3$, the accepted remaining-block-size theorem excludes the system: its forest bound leaves only $(3, 4), (4, 3)$ and its parity argument excludes both. These are all possible block systems. Thus every transitive alternative is impossible.

Finally the acquired exact early-cut calculation says that colour 3 has a unique eligible edge if and only if B is transitive on E. Each colour has at least one eligible edge, since its full omission is intransitive and G is transitive. The present exclusion therefore gives at least two. □

Attribution and scope. This is an assembly of the complete accepted primitive bound, pure-head obstruction and three newly reviewed imprimitive proofs. Their source attributions, retained inputs, full common theories, programs and distinct reviews remain paid. No new induction hypothesis is used inside the transitive-B exclusion; the induction that established the critical configuration remains explicitly inherited. This closes the transitive-B branch only. Opposite heads for $n \geq 26$, the unshared singleton endpoint, the odd-degree separator refinement and the full L044 recovery remain outstanding. No whole-original, formal or question credit is claimed.